CHAPTER 3B-DIFFERENT TOOLS TO CREATE JAVA PROGRAM

3b.1 Objective

- Compile and run program using javac and java
- Eclipse setup
- Using compile online website

3b.2 Content

3b.2.1 Download & Install JDK:

Oracle.com hosts the latest JDK installer and also the archives of older versions of JDK installer. The latest installer could be downloaded from the link:

http://www.oracle.com/technetwork/java/javase/downloads/index.html

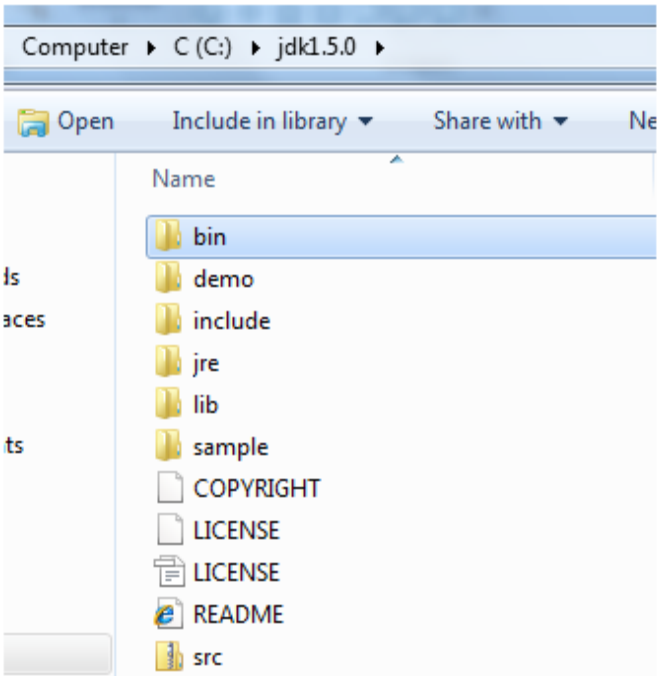
If you need a previous version of Java visit the following link:

http://www.oracle.com/technetwork/java/javase/archive-139210.html

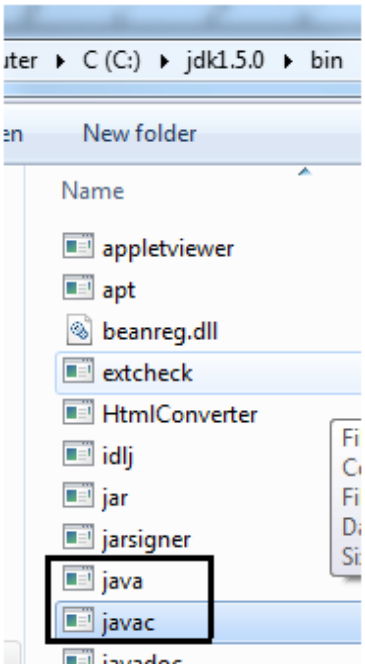
After the download is complete, double click the installer and follow the installation process. Follow the recommended settings during installation.

Depending on the location that was chosen to install the JDK, you would see the following contents. The outer most folder that has all the installed contents is generally referred to as Java Home.

So, as the per the below installation, the Java Home directory is C:\jdk1.5.0



The developer frequently needs an access to the java compiler (javac) and java interpreter (java). These tools come along with JDK and you will find them under the bin folder in Java installation folder.



To enable to developer to use these tools from any directory in the command prompt, we set these values in the operating system environment variable.

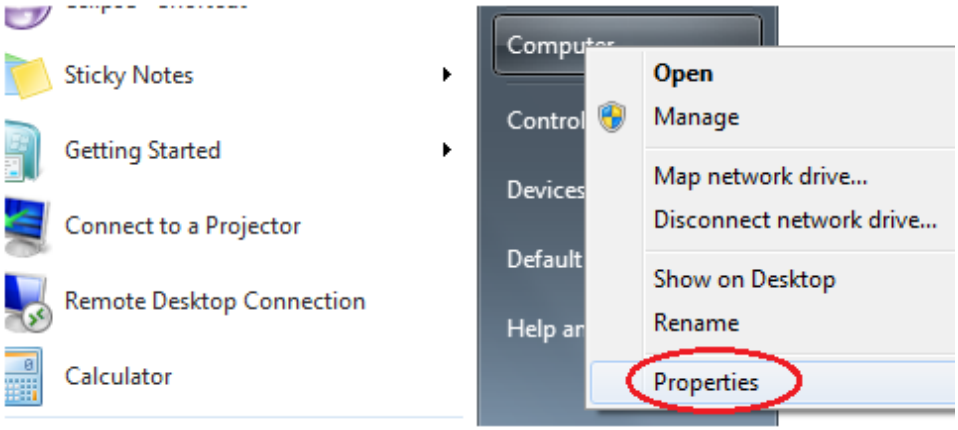
Configure JDK:

Environment variables are key and value pair. PATH is one of the environment variable (key). It's value is a collection of more than 1 directory path. Operating system uses the PATH variable to scan all the directory when a user want to execute a tool without mentioning the installation path.

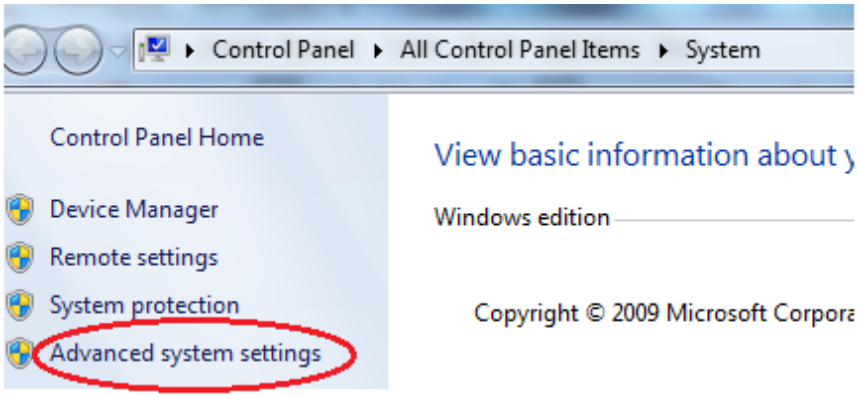
eg. PATH = C:\jdk1.5.0\bin; C:\Program Files\Intel\iCLS Client\; C:\Program Files\IBM\Trace Facility\

As you can see, the delimeter (;) is used to separate multiple directory mentioned as a value to the PATH environment variable.

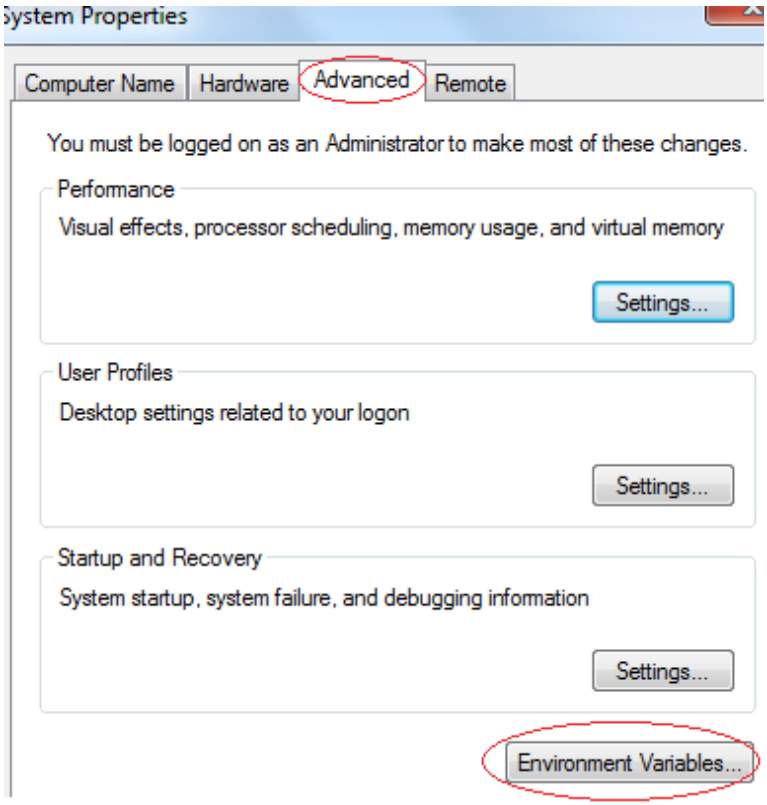
To set and environment variable, right click on Computer icon, and select Properties.



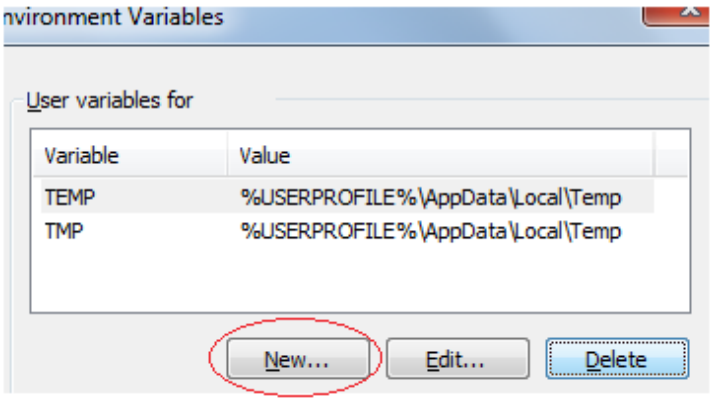
It will open Control Panel Home as shown below:



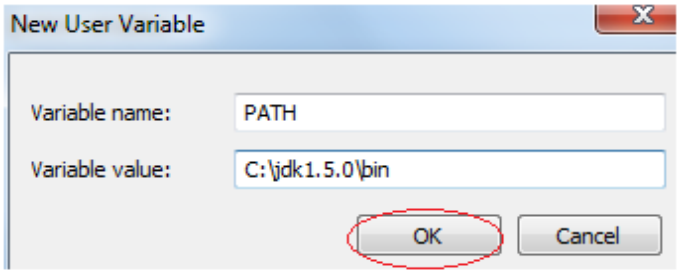
Right click on the Advanced system setting, it will open System Properties as shown below.



Go to the Advanced tab, and click on the Environment Variables button available at the bottom. This is would open a small window that will list all the existing environment variables. It has 3 button – New (to add a new environment variable), Edit (to edit the value of existing environment variable) and Delete (to delete and existing environment variable)

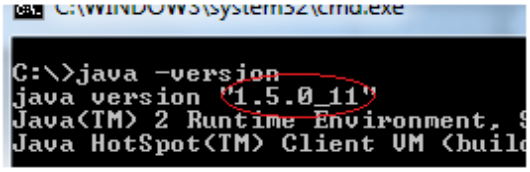


Click on the button – New. It will open a new window with 2 empty text boxes. Enter the variable name as PATH. In variable value, enter the path to the bin folder in java installation directory.



Click on OK to save changes.

You can verify the installation, by the command: java -version



Example:

Create a folder javaproject in your D: drive. Create a file MyFirstJavaProgram.java with following contents.

```
public class MyFirstJavaProgram
{
        public static void main(String a[])
        {
                System.out.println("Hello World");
        }
}
```

Open command prompt and browse to your folder D:\javaproject
Compile the file MyFirstJavaProgram.java with the help of the compiler – javac. The sytanx to compile a file with java extension is – **javac FileName.java**

If the compilation is successful, a file MyFirstJavaProgram.class is generated. The generated class file

can be run with help of an interpreter – java. The syntax to execute a class file is – **java FileName**

```
D:\javaproject>java MyFirstJavaProgram
Hello World

D:\javaproject>_
```

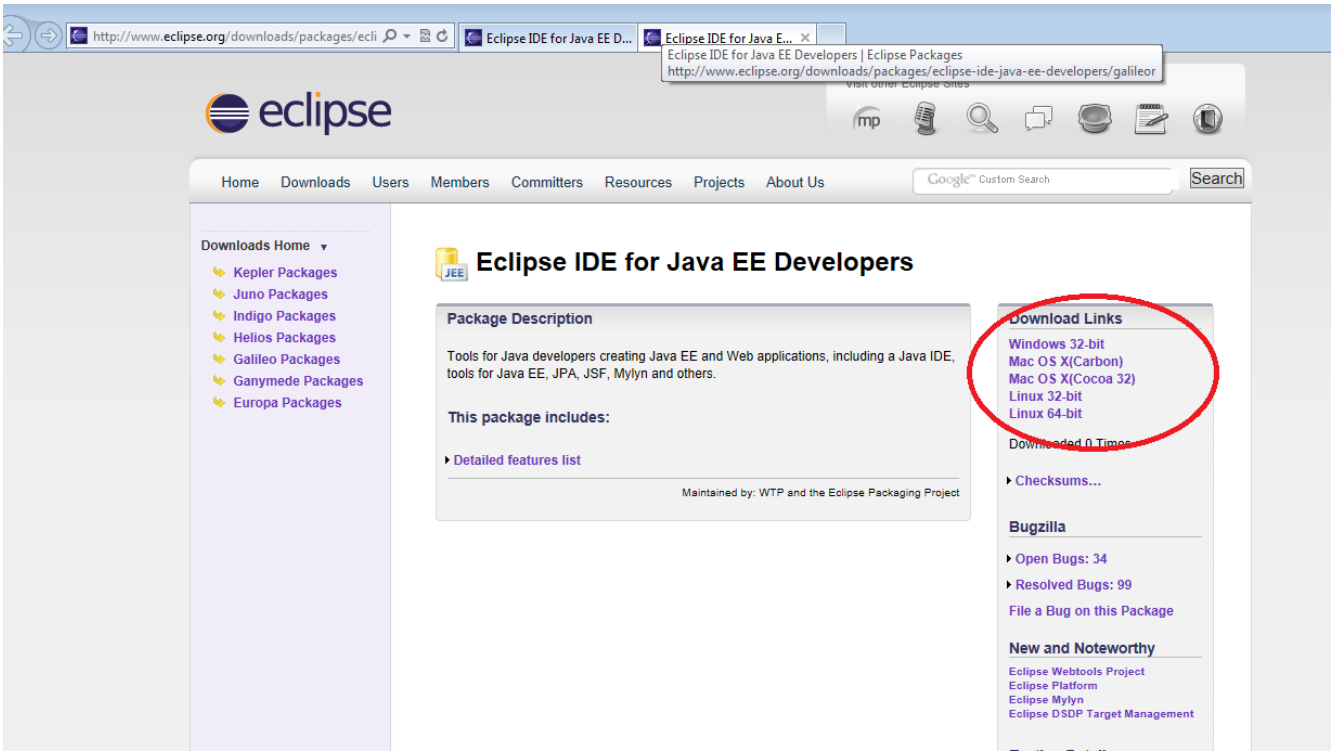
Congratulations !! You have executed your first java program.

3b.2.2 Eclipse

Good as of now we have learnt how to install JDK on our systems. We also learnt how to write a java program, compile it and run the java program. Let us now learn another way of writing Java program. That is using an IDE. IDE means Integrated Development Environment. There are many IDE's available in the market but we will be using Eclipse. Eclipse is an Integrated Environment which provides programmers perform a lot of activities required during development at one place. Some of the activities that can be done using Eclipse are writing code, making code more readable, like indentation, providing spacing, compiling, running, debugging etc. There are many tools available in the market which helps in doing each of the activity mentioned above in separate windows, but IDE integrates all the above windows into one environment. Using Eclipse we can write programs in many languages like C, CPP and Java, html, Perl etc. But we will only be focused on learning how to use Eclipse to develop Java programs. First of all let us learn how to get eclipse installed in our system.

Install Eclipse:

Eclipse has been evolving and there are many versions available and you may choose any of the existing versions for java development but we would recommend use of Galileo. You can download the same using the following url http://www.eclipse.org/downloads/packages/eclipse-ide-java-ee-developers/galileor . You may download the appropriate version depending upon your Operating System. Please refer the below diagram for the same.



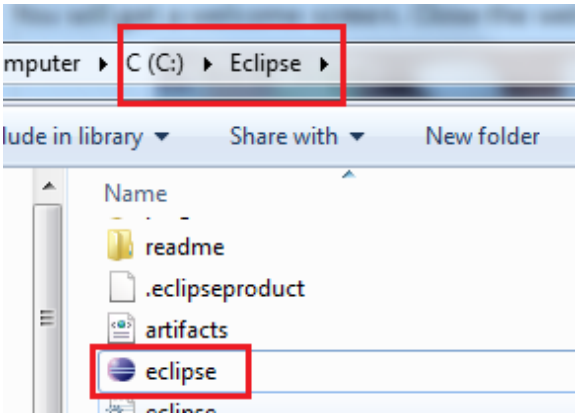
Make sure you download the right installation file based on your OS (winodws, linux or Mac) and architecture (32 or 64 bit). You may download and place the archive (zip, tar.gz) at any location of the hard disk of your system. Then the archive needs to be extracted using any of the archiving tools provided by your OS. The installation of eclipse is done just by unzipping the downloaded zip file.

Configure Eclipse:

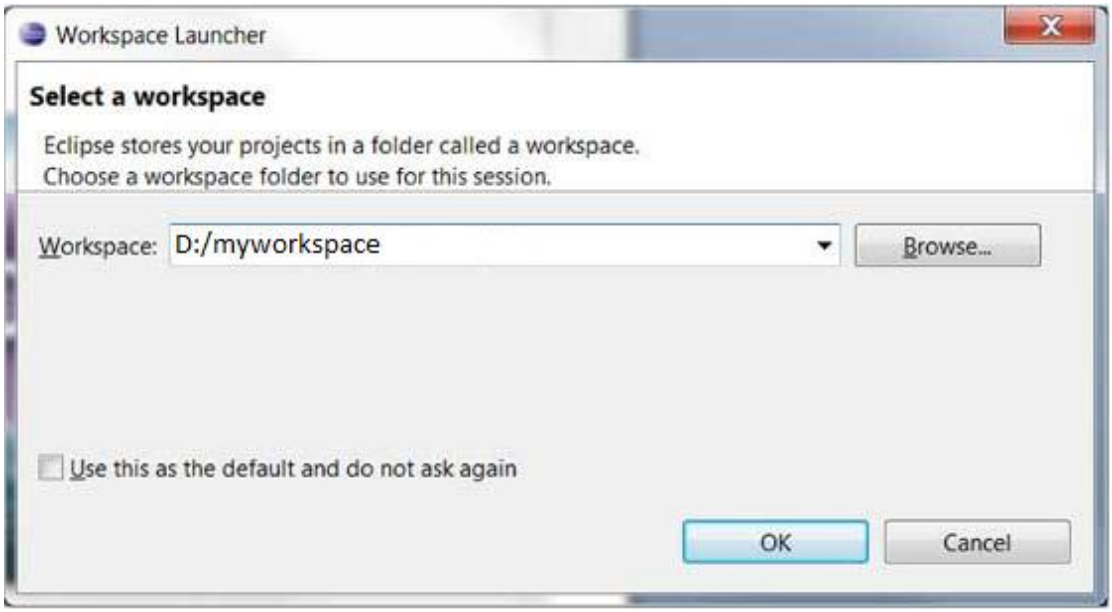
Before starting eclipse, make sure that you have downloaded Java Development Kit(JDK). We recommend you to download version 1.5. Refer to the download, install & configure JDK section present in this document for details.

Start eclipse from the eclipse installation directory. The eclipse installation directory will have an eclipse application as highlighted below.

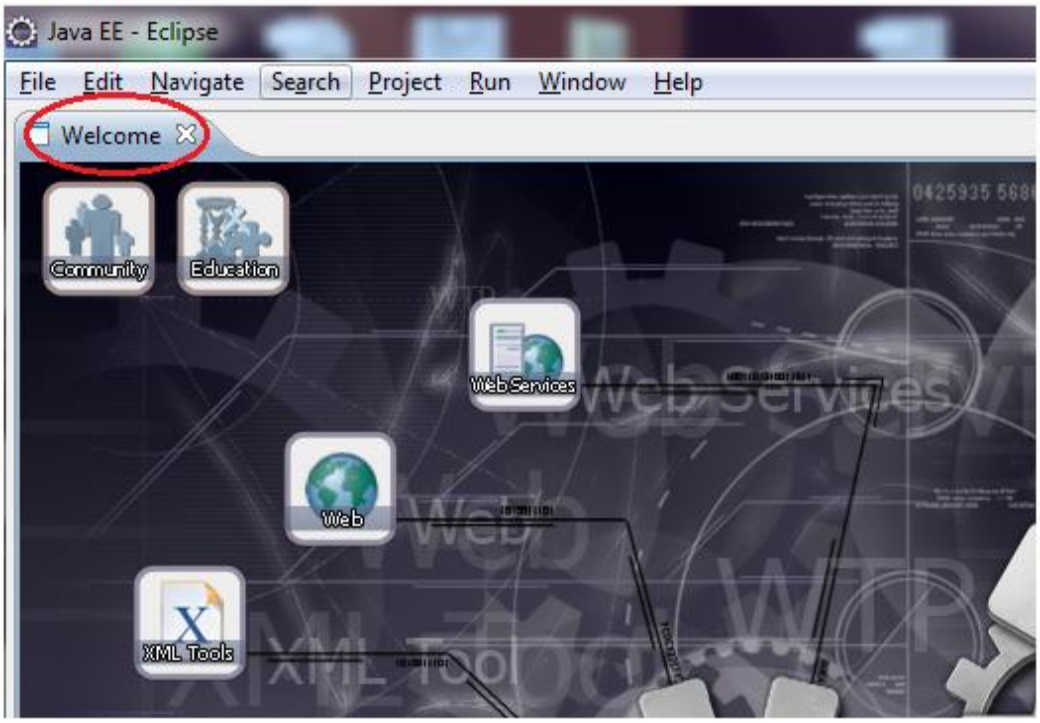
1. Start Eclipse



When eclipse starts up for the first time it prompts you for the location of the workspace folder. All your data will be stored in the workspace folder. You can accept the default or choose a new location. If you check "Use this as the default and do not ask again", this dialog box will not come up again.



After selecting workspace, you will see a welcome screen. Close the welcome screen.

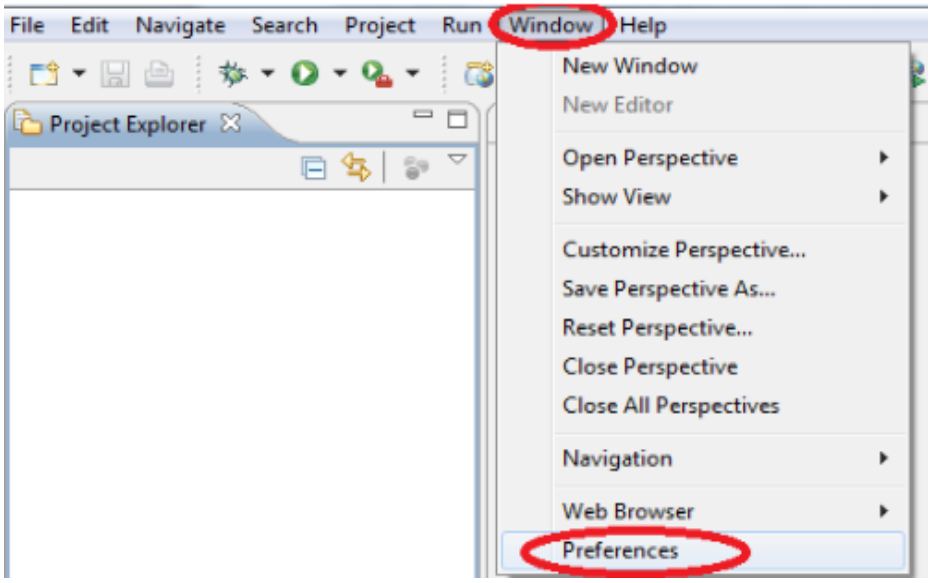


Setting up eclipse

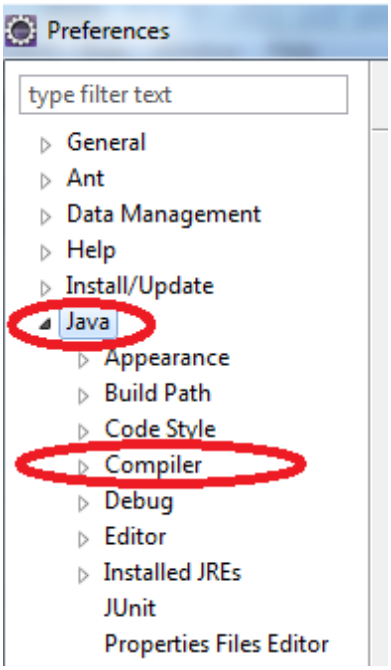
Now we need to make sure of certain things before proceeding with the creation of Java Project. We should set up environment for running programs in eclipse.

2.1 Compiler compliance level

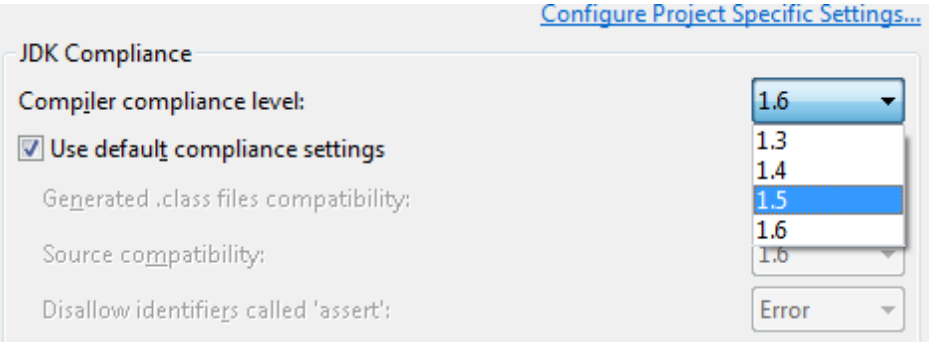
Right click Window and select Preferences.



Preferences window is opened. Now select Compiler from Java



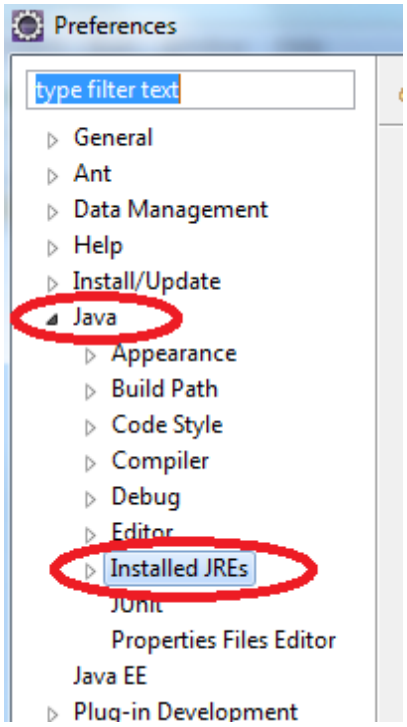
Compiler compliance level can be changed here. Select the appropriate version from drop down and click OK.



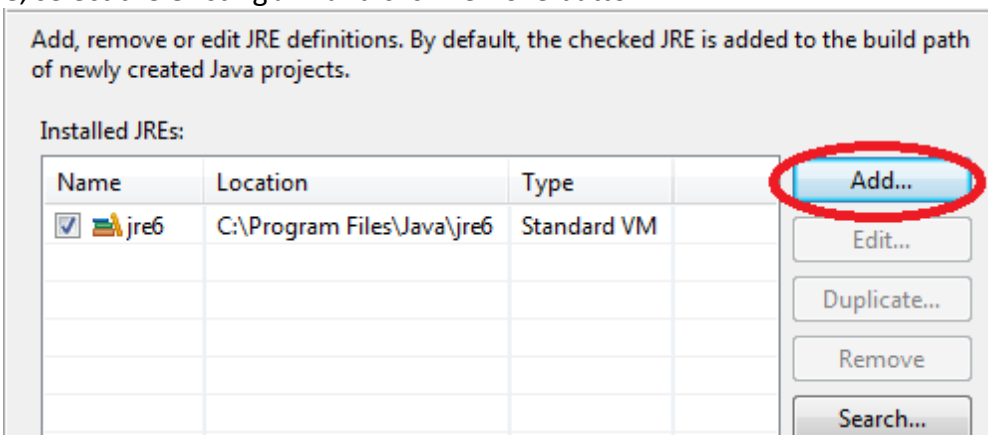
In the above scenario previous version was 1.6 which is now changed to 1.5

2.2 Installed JREs

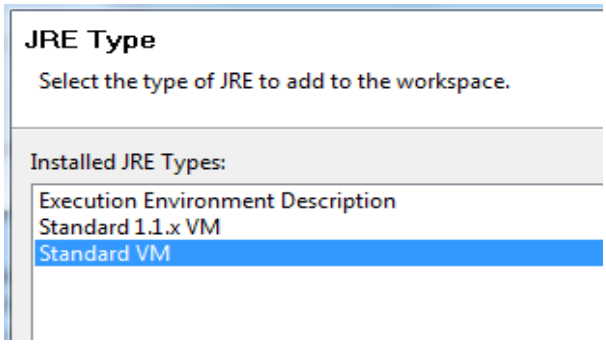
You can also add/remove JRE definitions using Installed JREs from Java of Preferences tab.



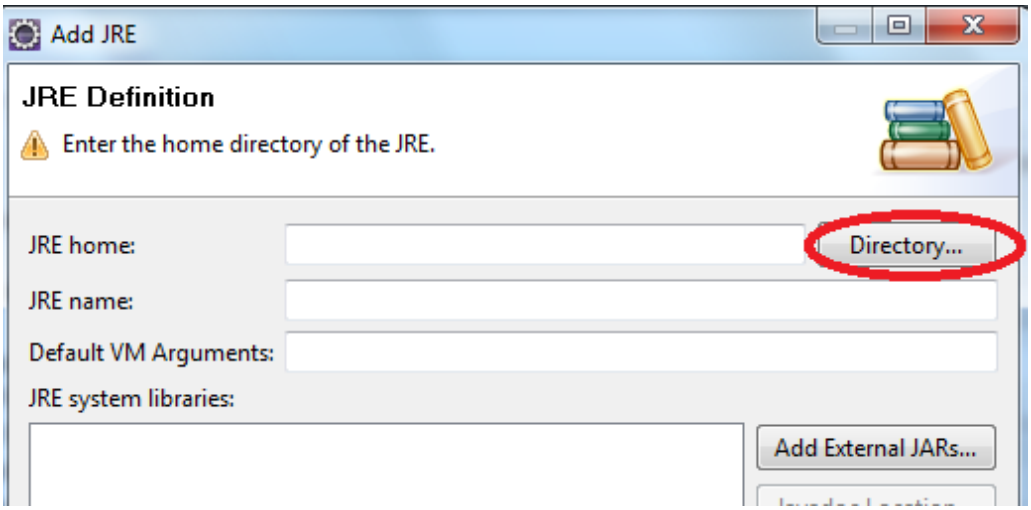
Now the default JRE is already checked in the Installed JREs, you can add new JRE using Add button and to remove, select the existing JRE and click Remove button.



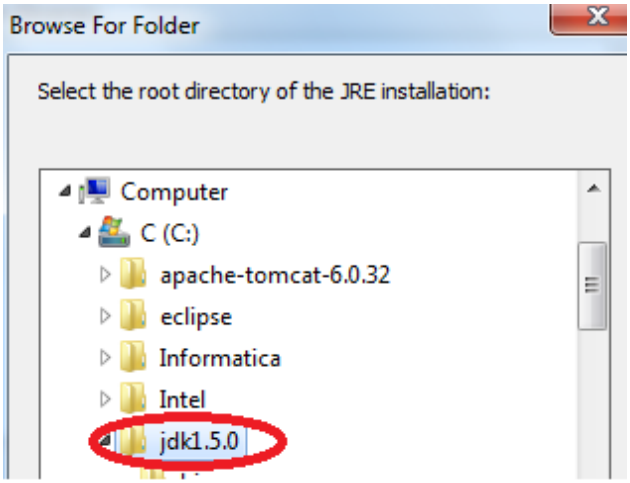
On clicking Add a window is opened where you need to select JRE Type



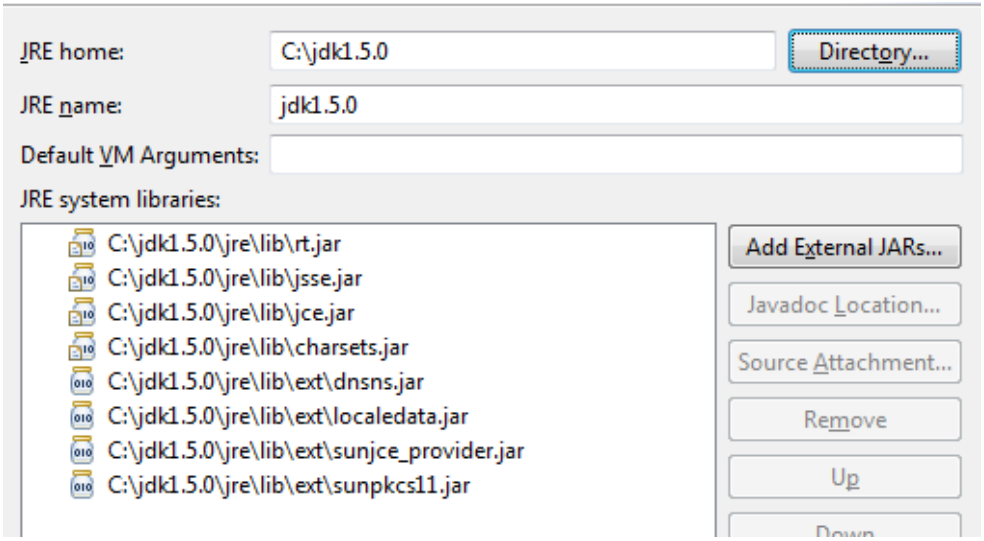
Select the type of JRE and click Next. Now you need to add JRE definition, for this you need to click on Directory.



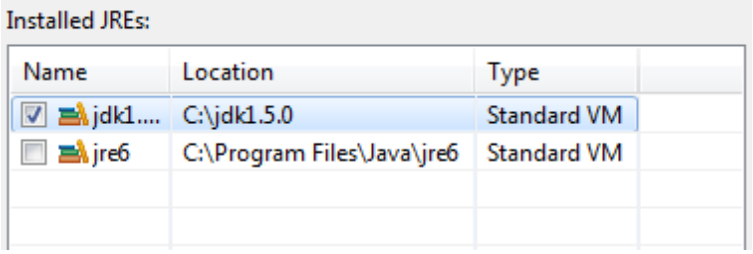
Choose the appropriate folder from the respective location and click OK



Now the library files related to the new definition are added. Click Finish to add it.



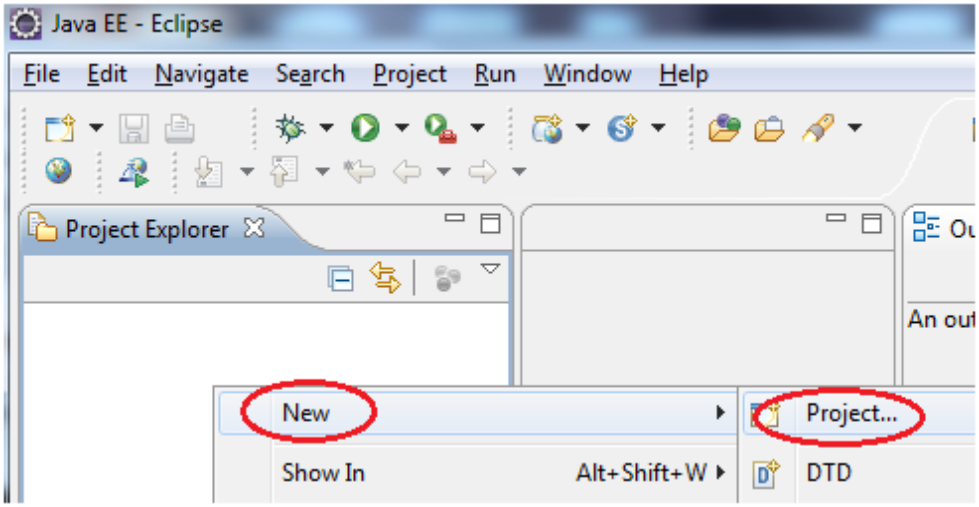
Select default JRE and click OK.



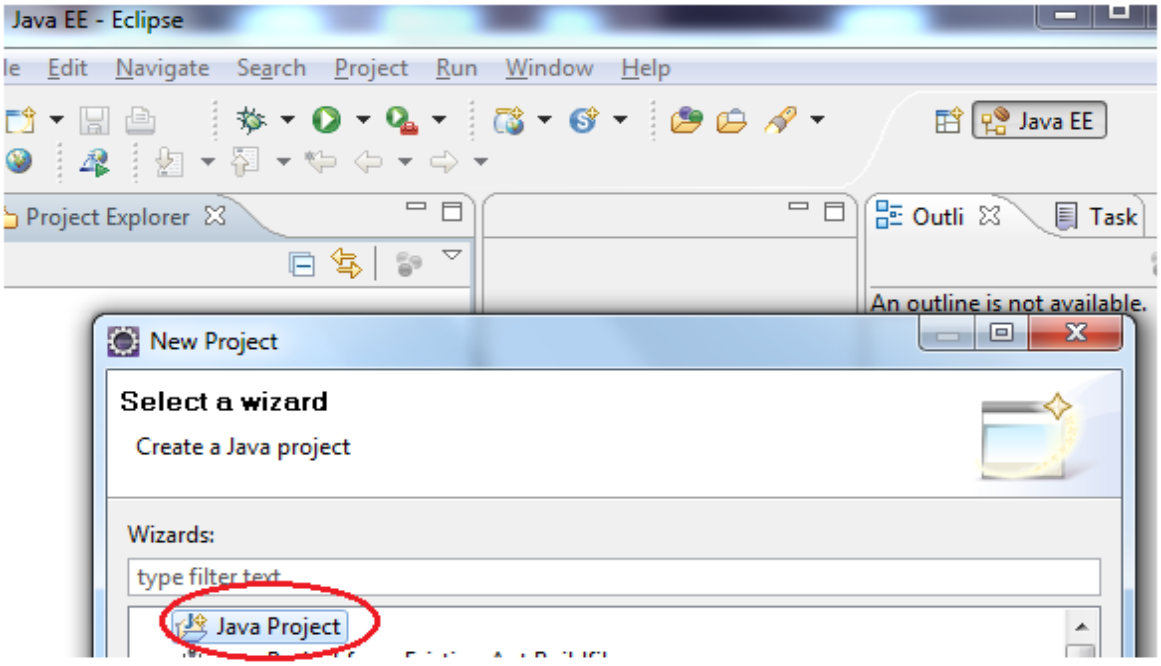
Now the setup is done.

3. Create Java Project

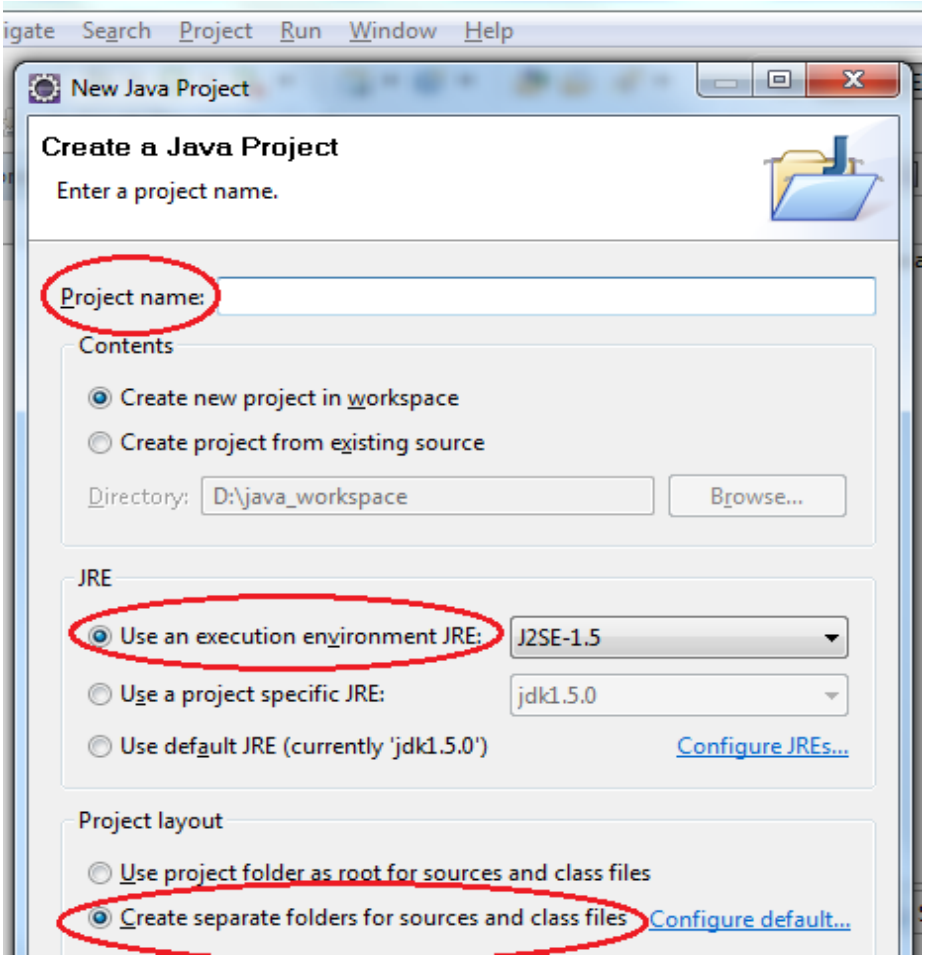
Right click anywhere in the Project Explorer area. Select New → Project.



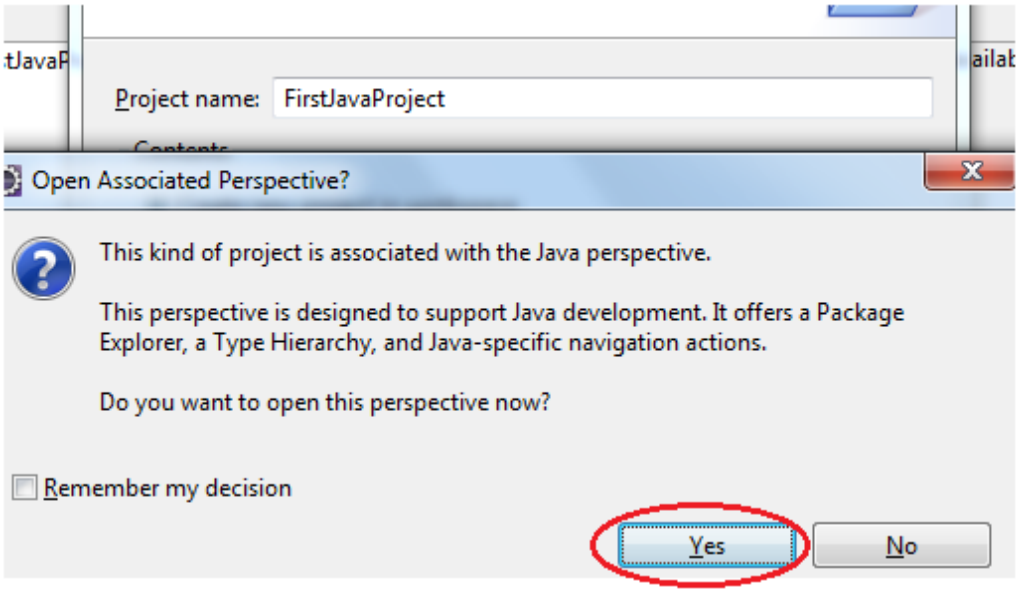
A wizard to create new project will appear. Select Java Project from the list of options available. Click Next.



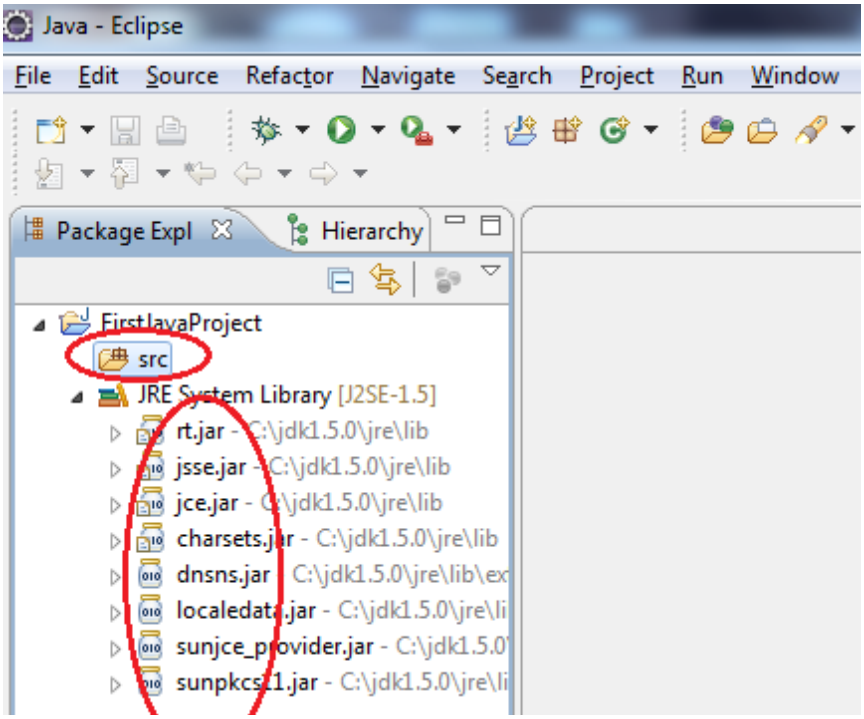
It opens a wizard to configure few setting for the new project. Enter the project name of your choice – eg. FirstJavaProject. Make sure your have select the below highlighted options. Click Finish.



Eclipse may ask you to open the associate perspective. Click Yes.

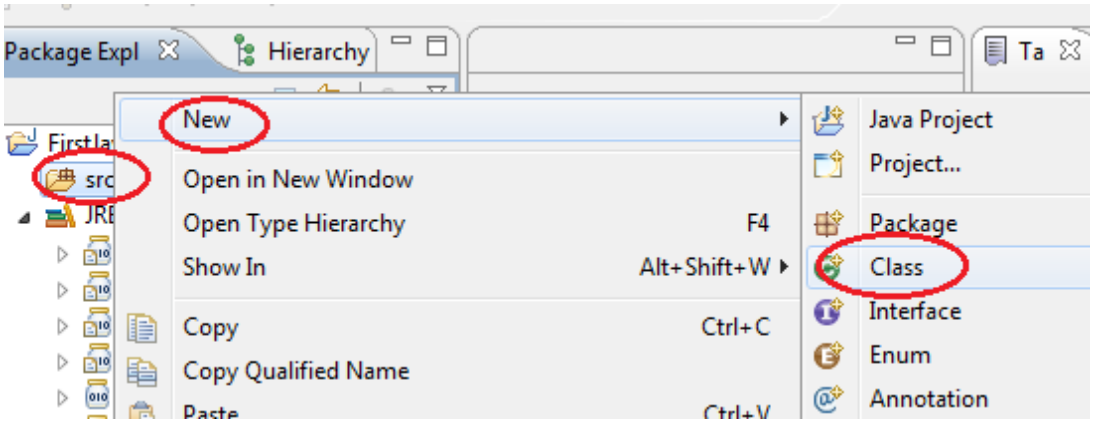


4. Project Directory.

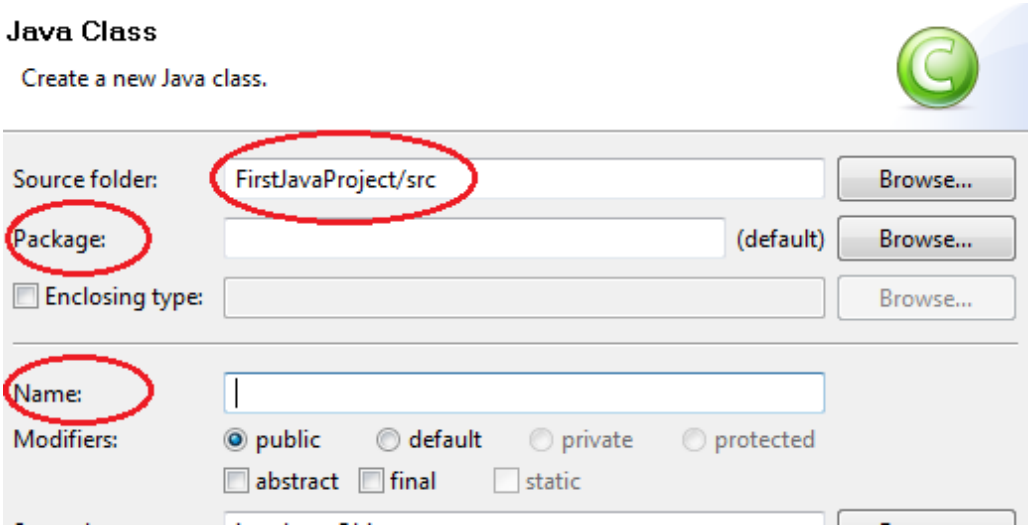


5. Create Java Class

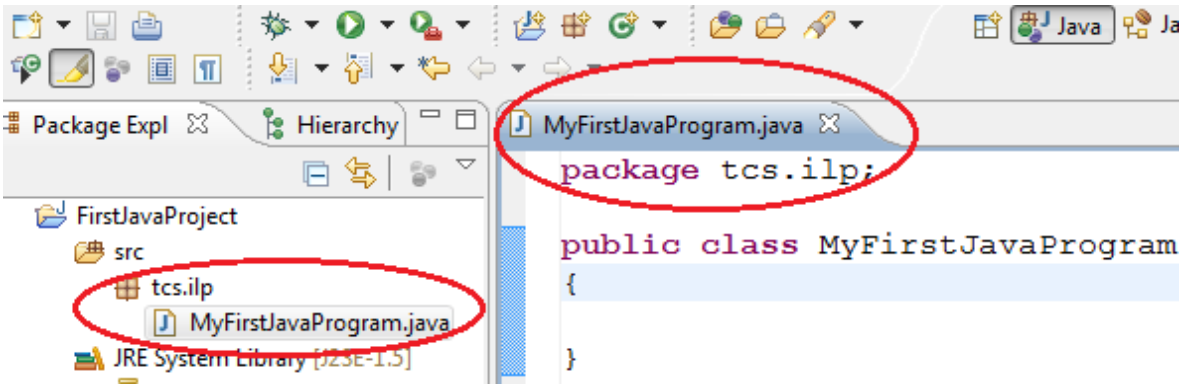
Right click on src folder. Select New → Class



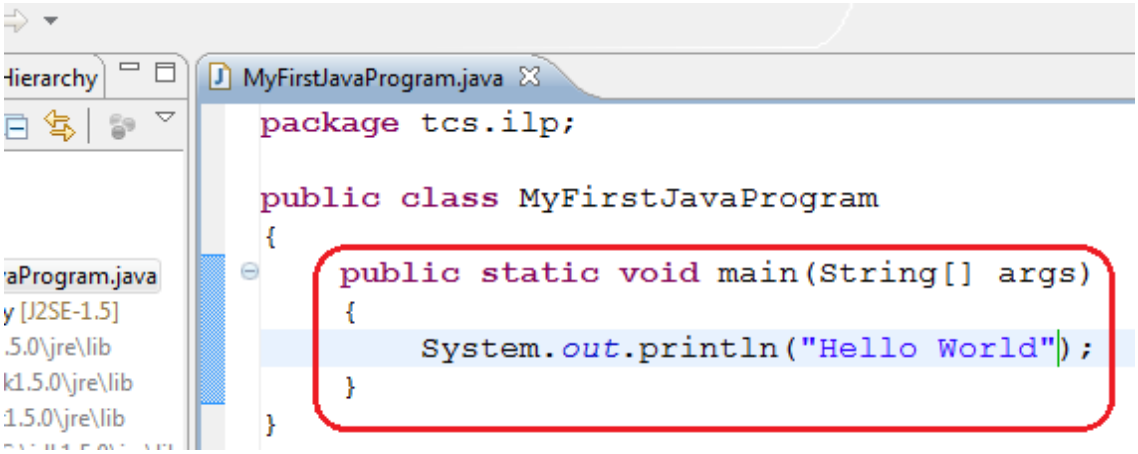
Reconfirm the location where you want the class to be created – its is in the src folder.
Enter the package name – **tcs.ilp**
Enter the class name - **MyFirstJavaProgram**



Eclipse created a package tcs.ilp under the src folder. It also creates a java file MyFristJavaProgram.java. Double click on the MyFristJavaProgram.java and eclipse opens the contents of the file in a window.

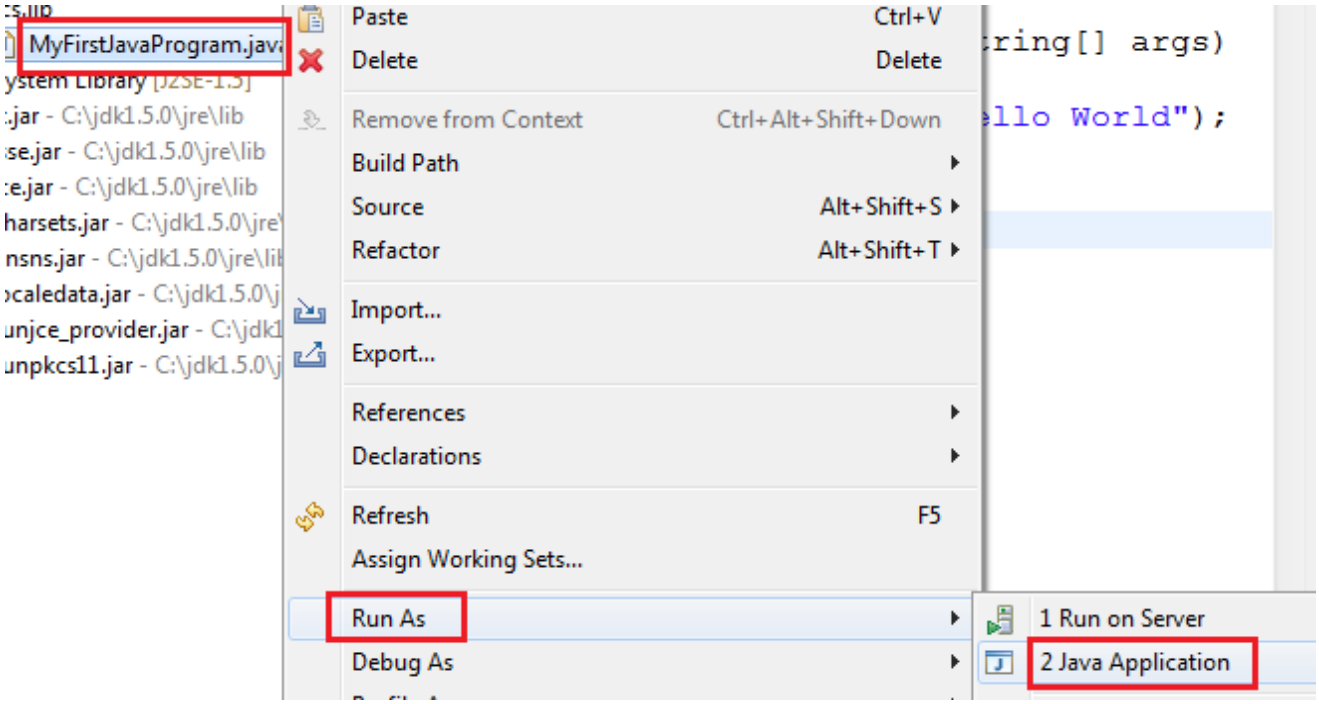


Add the main method and a simple print statement.

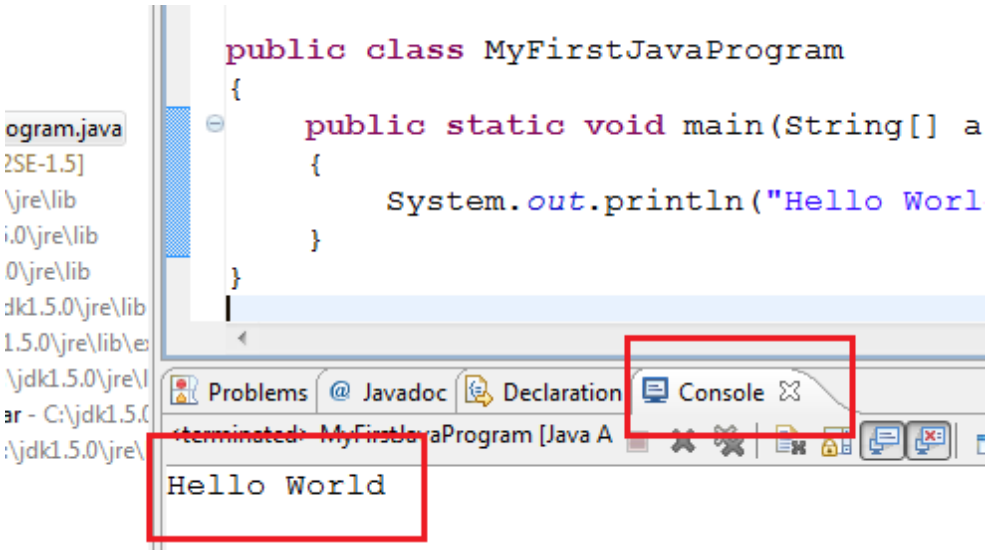


6. Execute the Java program

Right click on the file MyFirstJavaProgram. Select Run As → Java Application.



Here you do not need to compile the java file as eclipse does it for us automatically. We just need to run the program. Eclipse provides a console window where you can see the output of the program that you executed as Java Application.



3b.2.3 Compileonline:

compileonline.com is a web application which is used to compile and execute programming languages online. You can compile java programs as well. Here is the quick guide to compile and execute your program.

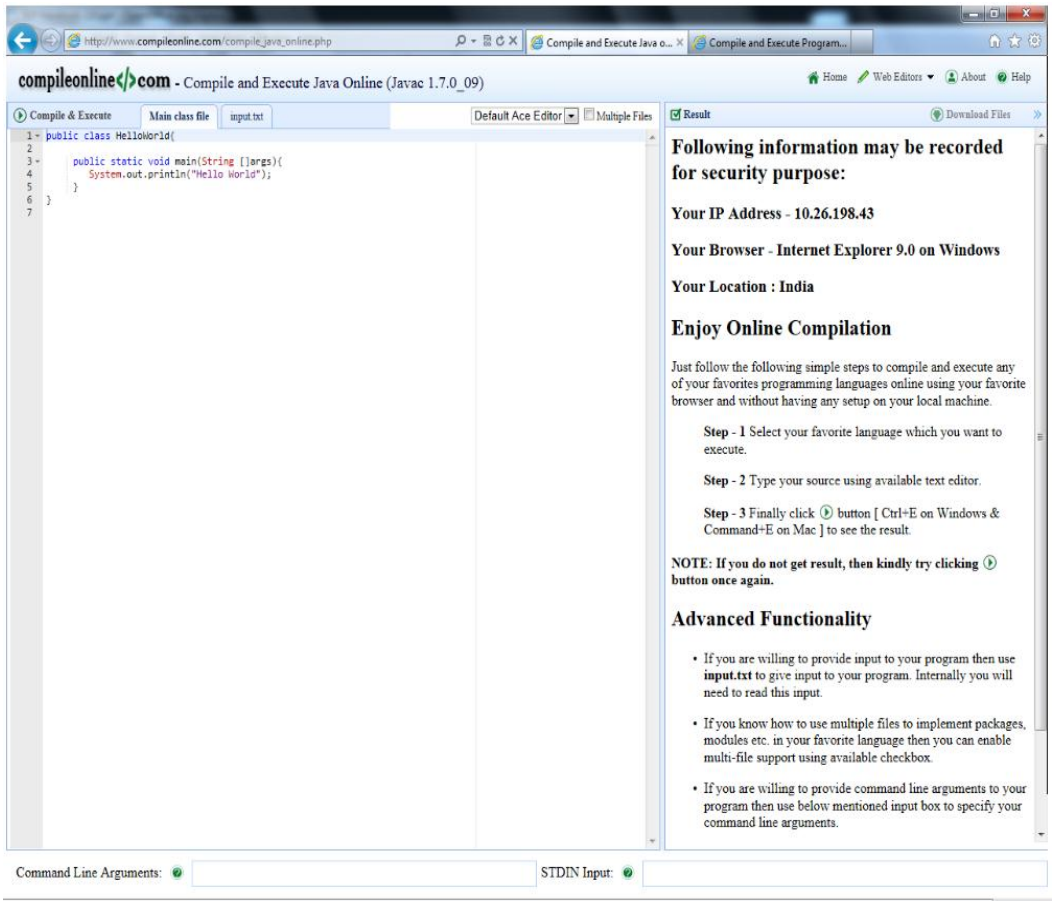
Step1:

Open browser and type www.compileonline.com **in the address bar.**

Step 2:

Click on "Java" in the Programming Languages section.

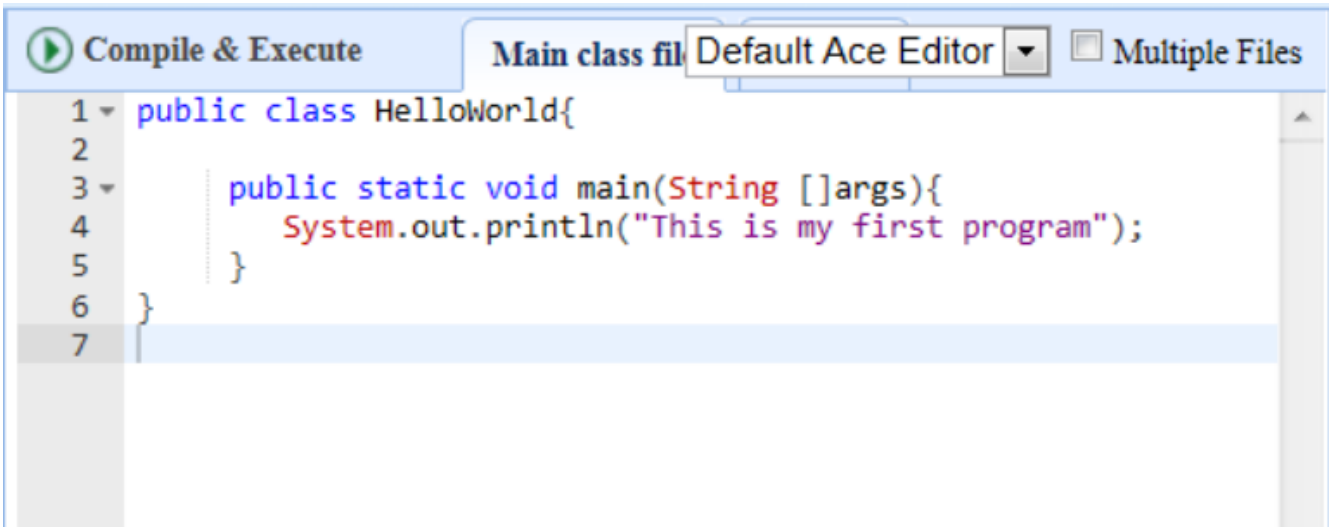
Now a window appears something like this:



The left hand area is where the source code is written and the right side “Result” tab is where the output is displayed. Steps to compile and execute the programs are given on the right hand area.

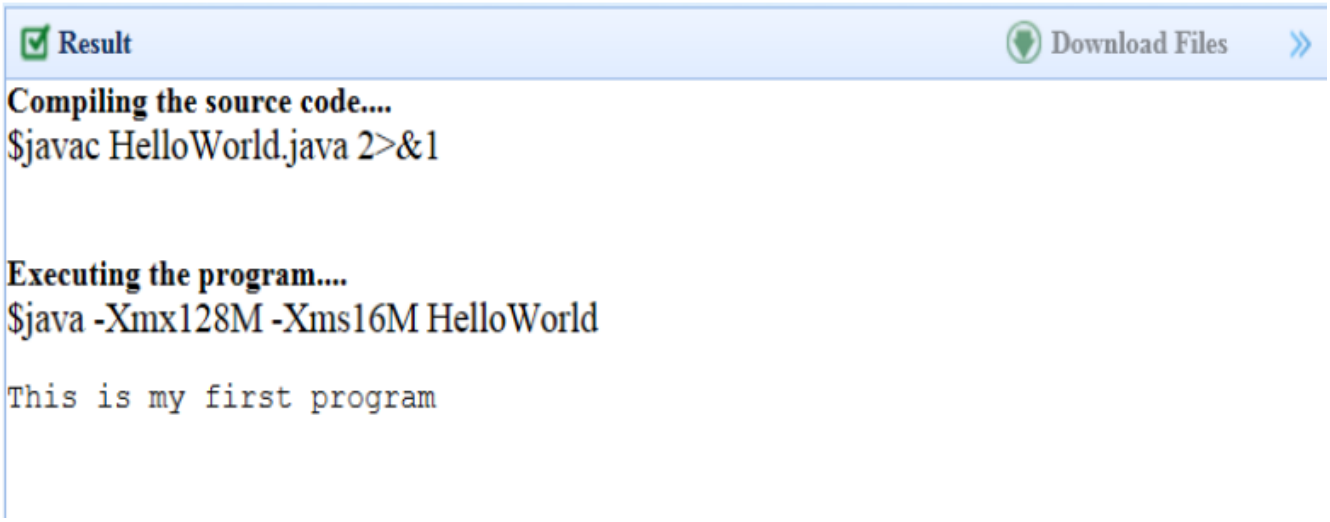
We will now execute a sample code.

Step 3: Write a program on the left hand area in “Main class file” tab



Step 4: Finally click the following button [Ctrl+E on Windows & Command+E on Mac] to see the result.

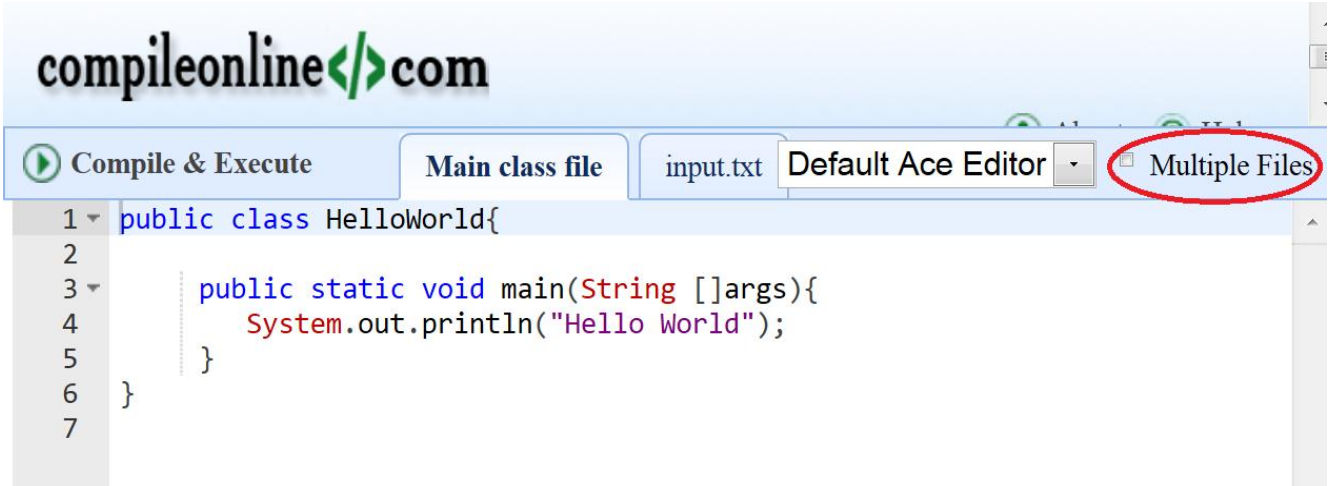
Step 5: The result is displayed on the right hand side



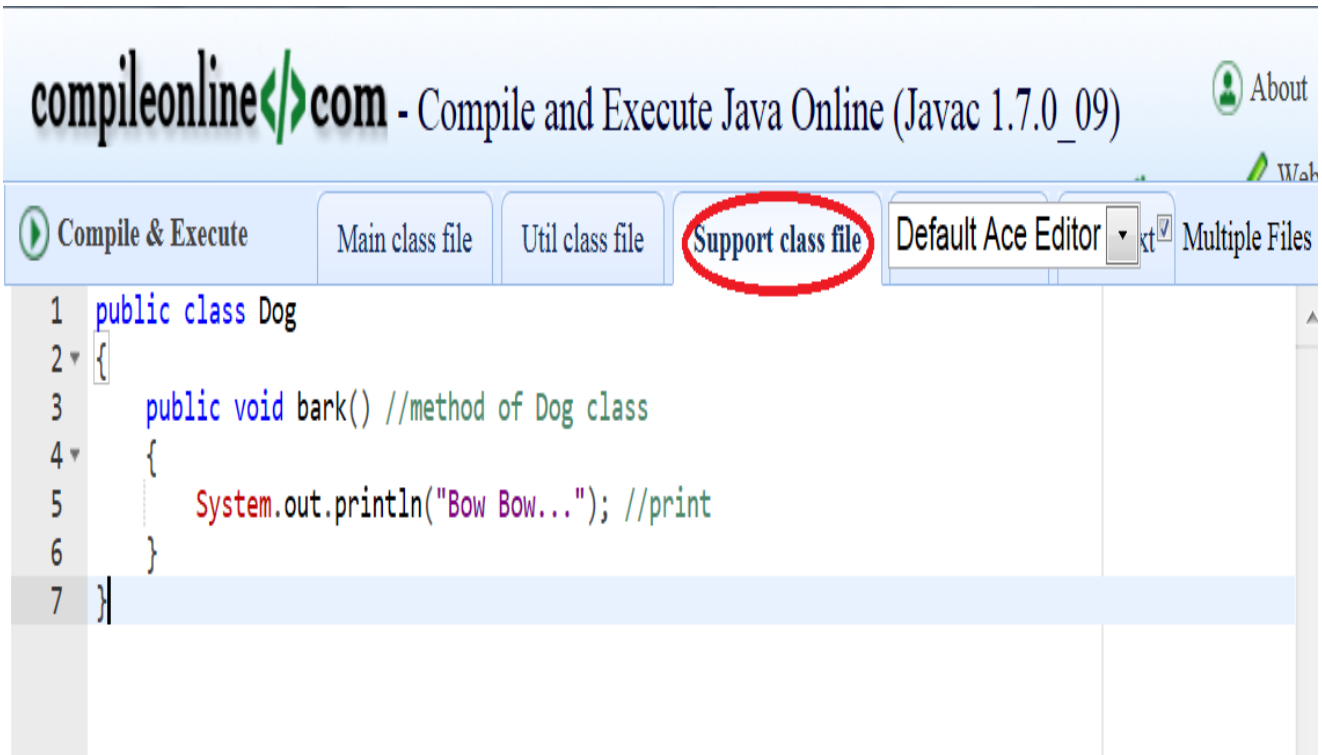
That's it!! Done!!

This application also helps us to execute multiple class files let us also see a small demo on how to execute multiple class files using compile online.

Step 1: Check the check box named "Multiple Files"



Step 2: Click "Support class file" and write the program in the area provided



In the above program, we have created a Dog class which has a method bark(). Now to

execute the above program, we need a Test class which contains main() method and a call to bark() method.

Step 3: Click Main class file tab and create a class with main() method

compileonline.com - Compile and Execute Java Online (Javac 1.7.0_09)

Compile & Execute | Main class file | Util class file | Support class file | Interface file | input.t

```
1  public class Test{
2
3      public static void main(String []args){
4        Dog d=new Dog(); //creating reference of type Dog
5        d.bark();    //calling method bark() which is in class Dog using reference
6      }
7  }
8
```

Step 4: Click Compile & Execute.

Result | Download Files

Compiling the source code....
$javac Test.java Dog.java 2>&1

Executing the program....
$java -Xmx128M -Xms16M Test Dog

```
Bow Bow...
```